Multi-Timeframe Key Levels: A Pre-Registered Replication and Null Result

Key-Level Research Project

2026-11-02 14:40 UTC

Abstract

We test a widely circulated multi-timeframe key-level strategy on eight liquid instruments over 2012–2025. On the 2012–2019 development sample (7,891 trades), the gross mean return is -0.015R per trade and the net mean return is -0.205R after a conservative half-spread-plus-slippage assumption (net annualised Sharpe -3.52). All 18 pre-registered parameter combinations have a negative net Sharpe ratio. On the sealed 2023–2025 holdout (3,155 trades), the net mean return is -0.166R and the net annualised Sharpe is -2.79; the pre-registered success criterion is not met. In a separate event study of executed signals, the strict hold rate (no close through the level over the next 96 bars) was 15.5%, and price reached +1R before -1R in 49.2% of cases. Even the weaker +1R measure is far below the 90% success rate claimed by practitioners. The evidence rejects the strategy as operationalised here: the gross mean return is slightly negative, the result is stable across a pre-registered parameter grid, and the sealed holdout does not reverse the conclusion.

Keywords: technical analysis; support and resistance; foreign exchange; intraday; backtest overfitting; replication

1 Introduction

Technical analysis remains widely used in foreign exchange and futures markets, yet the profitability of simple chart patterns after realistic costs and data-snooping corrections is contested [Brock et al., 1992, Lo et al., 2000, Menkhoff and Taylor, 2007, Osler, 2003]. This paper studies a specific and widely taught workflow: mark monthly, weekly, and daily support/resistance from candle bodies; determine the trend on the 4-hour chart; wait for a 15-minute rejection at a level; enter in the trend direction with a stop beyond the level and a target at the next level. The practitioner claim is that the level holds roughly 90% of the time. We translate the workflow into deterministic rules, freeze the rules before opening a sealed 2023–2025 holdout, and report the result whatever its sign. Our contribution is a well-powered negative result with three features that are often missing from practitioner replications: (i) intraday data from an independent vendor over 2012–2025, (ii) a pre-registered parameter grid and sealed holdout, and (iii) inference that corrects for the selection of the best variant [Bailey and López de Prado, 2014, Bailey et al., 2017, White, 2000, Hansen, 2005].

2 Related literature

The modern literature on technical trading begins with Brock et al. [1992], who documented predictability using simple rules on the Dow, and Sullivan et al. [1999], who showed that data-snooping can explain much of the apparent profitability. Lo et al. [2000] formalised pattern recognition and found some incremental information in chart patterns. In foreign exchange, Osler [2003] linked the predictive success of technical analysis to the clustering of stop and take-profit orders around round numbers, and Menkhoff and Taylor [2007] documented the continued

use of technical analysis by professionals. More recent work continues to find horizon- and regime-dependent predictability [Rink, 2023, Deprez and Frömmel, 2024]. Our design follows the backtesting protocol of Arnott et al. [2019] and the multiple-testing corrections of Harvey and Liu [2020].

3 Data

The primary sample is HistData.com Generic ASCII M1 bid bars for eight instruments: EURUSD, GBPUSD, USDJPY, AUDUSD, USDCAD, XAUUSD, SPXUSD, and NSXUSD, from 2012-01-01 to 2025-12-31. M1 bars are resampled into 15-minute, 4-hour, daily, weekly, and monthly bars. No future bar is used before its close. HistData timestamps are treated as fixed UTC-5 without daylight saving, per the vendor specification. The processed sample contains 346,372 EURUSD 15-minute bars, 330,920 XAUUSD bars, and 324,578 SPXUSD bars. An independent check compares weekly close-to-close returns with Yahoo Finance daily bars. The Yahoo series are not mixed into the historical backtest. The 2026 live shadow uses Yahoo 15-minute bars, whose public window is roughly 59 days, and is descriptive only.

Table 1: Instrument universe and base round-trip cost assumptions.

Instrument	Class	Half-spread	Slippage	Round trip
EURUSD	fx major	3e-05	2e-05	0.0001
GBPUSD	fx major	4e-05	3e-05	0.00014
USDJPY	fx major	0.003	0.002	0.01
AUDUSD	fx major	4e-05	3e-05	0.00014
USDCAD	fx major	4e-05	3e-05	0.00014
XAUUSD	metal	0.15	0.1	0.5
SPXUSD	equity index	0.25	0.25	1
NSXUSD	equity index	0.5	0.5	2

4 Strategy and execution

Monthly, weekly, and daily pivots are clustered in price with a tolerance of 0.50× ATR14(D1) and become tradable after at least two pivots. A level is a resistance if a 15-minute bar trades through it and closes back below; a support is the mirror case. The 4-hour trend is defined by the last two confirmed swing highs and lows. A trade requires trend agreement, a level touch, two consecutive closes on the rejected side, a directional candle, and a chase of no more than 1× ATR15. Entry is at the next 15-minute open. The stop is the more conservative of the level plus 0.25× ATR15 and the five-bar extreme plus 0.10× ATR15, capped at 4× ATR15. The target is the nearest opposite active level at least $1R$ away, defaulting to $2R$ and capped at $5R$. Maximum holding period is 96 bars. If the stop and target share a bar, the stop is assumed first. Risk is 0.5% of equity per trade. Costs are half the quoted spread plus adverse slippage on each side. A gap through the stop fills at the next available open; a gap through the target does not enter the trade.

5 Statistical protocol

The sample is split into development (2012–2019), validation (2020–2022), and a sealed holdout (2023–2025). The base specification is fixed by faithfulness to the transcript, not optimised. The 18-point grid is selected on development only; the winner is evaluated unchanged on validation and holdout. Inference uses the stationary bootstrap [Politis and Romano, 1994], the deflated

Table 2: Frozen base parameters.

Parameter	Value
entry timeframe	15min
trend timeframe	4h
level timeframes	['1D', '1W', '1M']
pivot left	2
pivot right	2
trend pivot left	3
trend pivot right	3
level price source	body
cluster atr	0.5
min touches	2
level expiry days	730
touch lookback	300
confirmation closes	2
max chase atr	1.0
stop buffer atr	0.25
stop extreme lookback	5
stop extreme buffer atr	0.1
max stop atr	4.0
default rr	2.0
max rr	5.0
max hold bars	96
risk per trade	0.005
require directional candle	True
trend method	swing
require trend	True
ema length	200
ema slope bars	5

Sharpe ratio [Bailey and López de Prado, 2014], the probability of backtest overfitting via CSCV [Bailey et al., 2017], the White Reality Check [White, 2000], Hansen's SPA test [Hansen, 2005], and Benjamini–Hochberg control of the per-instrument false discovery rate [Benjamini and Hochberg, 1995]. The holdout is opened once, after a SHA-256 code freeze.

6 Results

On the development sample the base strategy produced 7,891 trades. Gross mean return was -0.015R per trade and net mean return was -0.205R, with net annualised Sharpe -3.52 and maximum drawdown -99.97

Table 3: Base strategy by sample split.

Metric	Dev 2012–19	Val 2020–22	Holdout 2023–25	Val winner	Holdout winner
Trades	7,891	3,556	3,155	1,831	1,460
Trading days	2,087	783	783	783	783
Annual return	-62.38%	-53.72%	-55.89%	-39.54%	-25.05%
Annual volatility	26.72%	30.17%	27.91%	22.54%	19.93%
Net Sharpe	-3.52	-2.40	-2.79	-2.12	-1.35
Max drawdown	-99.97%	-91.62%	-92.45%	-80.51%	-59.83%
Win rate	29.07%	30.76%	29.38%	26.49%	27.19%
Mean net R	-0.205	-0.121	-0.166	-0.169	-0.141
Mean gross R	-0.015	+0.002	-0.028	-0.019	+0.025
Cost drag	+0.190	+0.123	+0.138	+0.150	+0.166
Profit factor	0.747	0.838	0.789	0.795	0.831
Stop rate	65.25%	64.26%	65.86%	69.52%	68.22%
Target rate	18.72%	21.46%	19.24%	18.79%	16.51%

Table 4 reports the full grid. Every combination has a negative net Sharpe. The least-negative variant is grid_c1.0_m3_k3 (Sharpe -2.26), selected only by ranking losses. Figure 2 shows the same surface graphically. On validation (2020–2022) the base strategy produced

Table 4: Pre-registered parameter grid on the development sample.

Variant	Cluster	Touches	Confirms	Trades	Sharpe	Mean net R	Win rate
grid_c1.0_m3_k3	1.00	3	3	3,675	-2.26	-0.215	25.66%
grid_c1.0_m3_k2	1.00	3	2	4,760	-2.46	-0.195	27.63%
grid_c1.0_m2_k3	1.00	2	3	4,494	-2.61	-0.218	26.10%
grid_c0.25_m3_k3	0.25	3	3	5,208	-2.69	-0.201	28.32%
grid_c1.0_m2_k2	1.00	2	2	5,831	-2.75	-0.191	28.18%
grid_c0.25_m3_k2	0.25	3	2	6,774	-2.78	-0.173	29.88%
grid_c1.0_m3_k1	1.00	3	1	7,191	-2.84	-0.174	29.15%
grid_c0.5_m3_k2	0.50	3	2	6,002	-2.94	-0.201	28.41%
grid_c0.25_m3_k1	0.25	3	1	10,071	-2.98	-0.156	31.30%
grid_c0.5_m3_k3	0.50	3	3	4,609	-3.09	-0.239	26.60%
grid_c1.0_m2_k1	1.00	2	1	8,702	-3.23	-0.177	29.56%
grid_c0.5_m3_k1	0.50	3	1	8,922	-3.33	-0.189	29.44%
grid_c0.5_m2_k2	0.50	2	2	7,891	-3.52	-0.205	29.07%
grid_c0.25_m2_k3	0.25	2	3	7,547	-3.60	-0.219	29.07%
grid_c0.5_m2_k3	0.50	2	3	6,073	-3.62	-0.242	27.27%
grid_c0.25_m2_k2	0.25	2	2	9,712	-3.75	-0.193	30.55%
grid_c0.25_m2_k1	0.25	2	1	14,328	-3.93	-0.169	32.01%
grid_c0.5_m2_k1	0.50	2	1	11,709	-3.98	-0.191	30.23%

3,556 trades with net mean R -0.121 and net Sharpe -2.40. The development grid winner produced 1,831 trades with net mean R -0.169. The sign did not reverse. On the sealed holdout (2023–2025) the strategy produced 3,155 trades with net mean R -0.166, net Sharpe -2.79, and

Table 5: Pre-registered ablations on the development sample.

Ablation	Trades	Sharpe	Mean net R	Win rate
no_trend	32,645	-11.01	-0.215	31.24%
no_confirmation	14,088	-4.52	-0.214	29.15%
no_directional	12,217	-4.69	-0.247	27.51%
ema_trend	11,566	-3.50	-0.191	27.36%
wick_levels	7,719	-2.79	-0.179	29.28%
min_touches_3	6,002	-2.94	-0.201	28.41%
cluster_025	9,712	-3.75	-0.193	30.55%
cluster_100	5,831	-2.75	-0.191	28.18%
confirmation_1	11,709	-3.98	-0.191	30.23%
confirmation_3	6,073	-3.62	-0.242	27.27%

maximum drawdown -92.45 All multiple-testing corrections agree with the negative verdict. The

Table 6: Holdout cost sensitivity.

Cost multiplier	Trades	Mean net R	Net Sharpe
0.00x	3,155	-0.028	-0.42
1.00x	3,155	-0.166	-2.79
2.00x	3,155	-0.304	-5.02
3.00x	3,155	-0.442	-7.05

Table 7: Multiple-testing inference on the holdout.

Test	Value	Detail
DSR (base)	1.14e-23	-0.18 vs threshold 0.20
DSR (grid winner)	4.17e-15	-0.08 vs threshold 0.20
PBO (CSCV)	0.1071	924 splits, 18 variants
Reality Check p-value	1.0000	stat -0.04
SPA p-value	1.0000	stat 0.00
Alpha vs equal-weight (base)	-54.46%	t = -5.09, p = 3.65e-07; beta 0.06
Alpha vs equal-weight (winner)	-23.93%	t = -2.52, p = 0.0119; beta 0.04
Bootstrap 95% CI, mean net R	[-0.228, -0.101]	point -0.166; 5,000 resamples
Bootstrap 95% CI, net Sharpe	[-3.96, -1.66]	point -2.79; annualised
Trials in DSR	28	grid + ablations
Random-entry placebo p-value	0.7180	observed -0.166 vs placebo -0.149
Random-level placebo p-value	¡0.0050 (0/200)	observed -0.166 vs placebo -0.239

deflated Sharpe ratio is essentially zero (1.14e-23 for the base and 4.17e-15 for the grid winner). The PBO is 0.11, but this is modest only because the ranking of a uniformly unprofitable set is stable; it is not evidence of a profitable variant. The stationary-bootstrap 95% confidence interval is [-0.228, -0.101] for mean net R and [-3.96, -1.66] for the annualised net Sharpe; both lie entirely below zero. The annualised alpha against an equal-weight portfolio of the same instruments is -54.46Table 8 tests the mechanism directly. It reports the share of executed development signals for which the marked level held and the share for which price reached $+1R$, $+2R$, or $+3R$ before the $-1R$ stop. The 90% claim would require the $+1R$ rate to be near 90%; it is not. Across 40 processed series and 2,930,707 bars, there were 0 duplicate timestamps, 0 OHLC violations, and 0 non-positive prices; the largest absolute 15-minute log return was 7.6Intrabar ambiguity was measured directly: 8 of 7,891 trades (0.1The 2026 live shadow produced 207 trades with mean net R -0.139 and win rate 30.92 Table 10 compares the strategy with simple alternatives on the same holdout window and cost model. Buy-and-hold and the 50/200-day SMA crossover are

Table 8: Level-reaction audit on executed development signals.

Level	Side	N	Hold rate	Hit +1R	Hit +2R	Hit +3R
1D	-1	3,262	15.79%	48.65%	31.55%	22.32%
1D	1	3,160	15.76%	50.47%	31.74%	21.93%
1M	-1	38	31.58%	52.63%	36.84%	28.95%
1M	1	44	6.82%	40.91%	18.18%	11.36%
1W	-1	696	13.22%	47.70%	31.03%	20.98%
1W	1	691	14.91%	48.34%	29.09%	20.41%
ALL	0	7,891	15.50%	49.25%	31.31%	21.85%

Table 9: Cross-vendor weekly-return validation against Yahoo Finance.

Symbol	Weeks	Return corr.	Mean abs weekly diff
EURUSD	730	0.79	0.53%
GBPUSD	730	0.77	0.58%
USDJPY	730	0.76	0.64%
AUDUSD	730	0.78	0.67%
USDCAD	730	0.76	0.50%
XAUUSD	730	0.98	0.31%
SPXUSD	729	1.00	0.14%
NSXUSD	729	1.00	0.14%

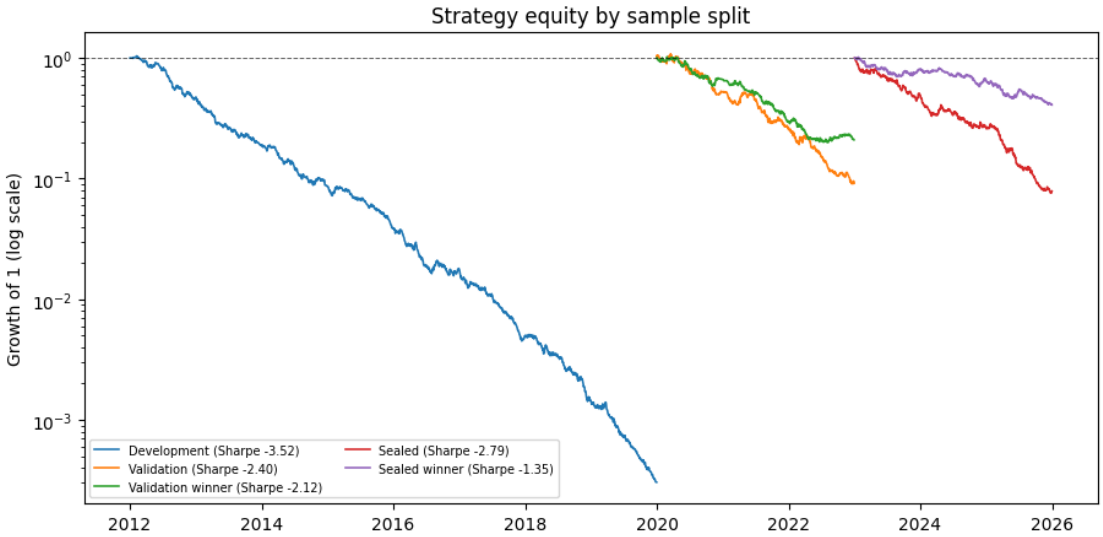


Figure 1: Strategy equity by sample split (log scale).

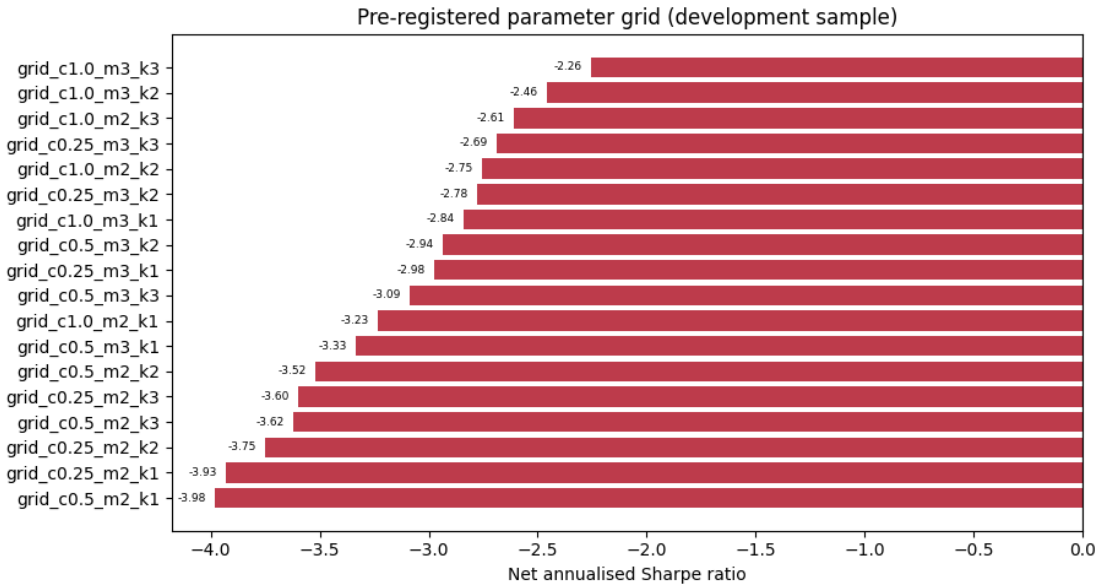


Figure 2: Pre-registered parameter grid (development).

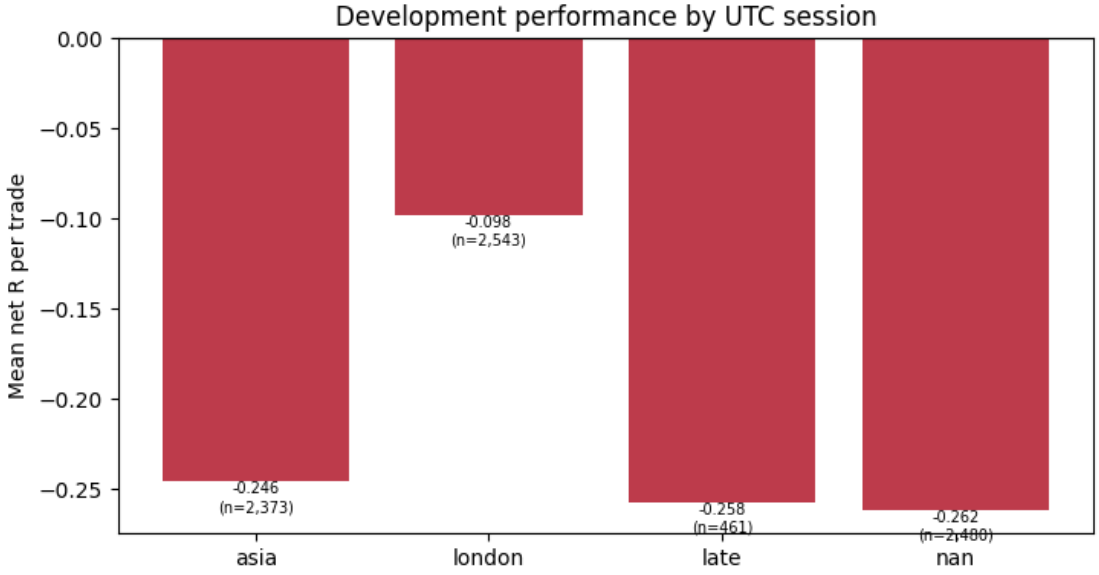


Figure 3: Development mean net R by UTC session.

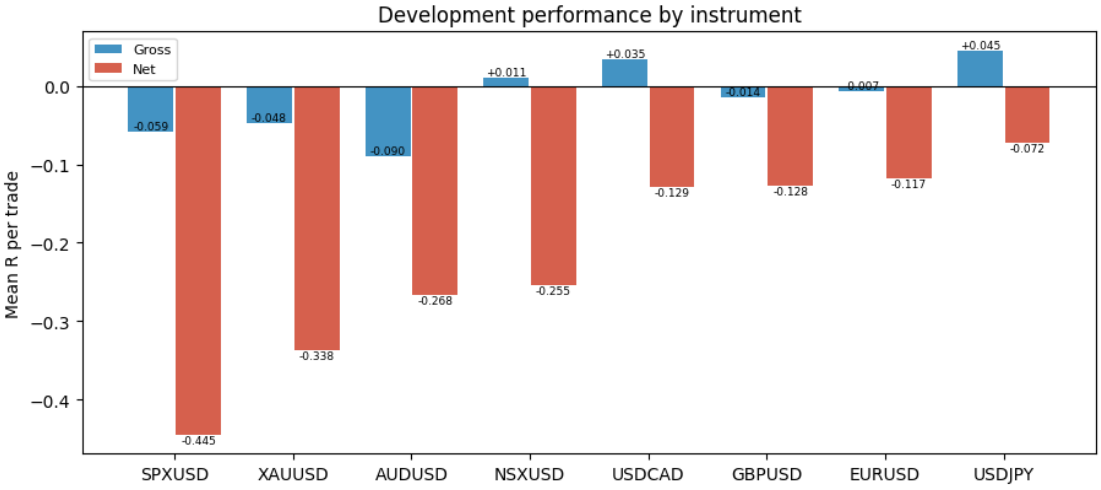


Figure 4: Development performance by instrument.

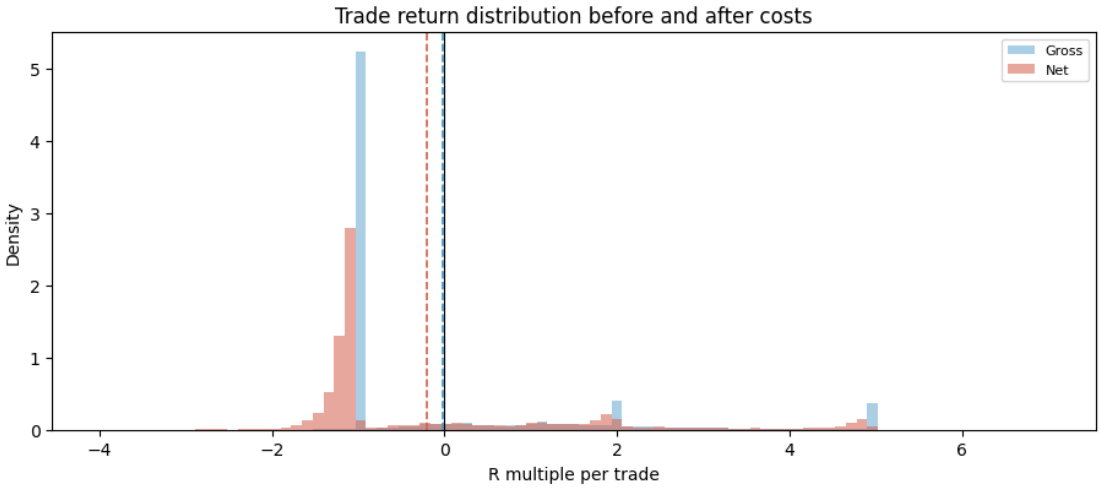


Figure 5: Trade return distribution before and after costs.

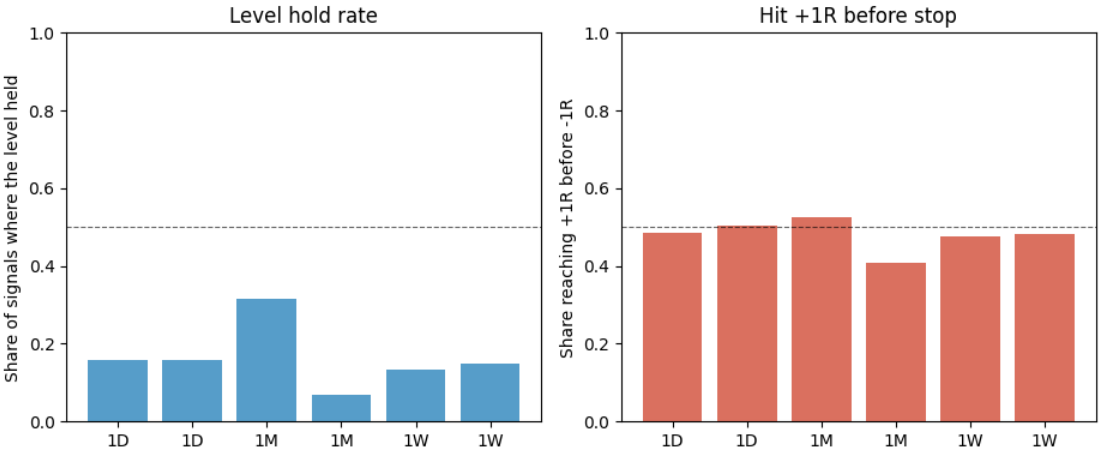


Figure 6: Level hold rate and +1R-before-stop rate.

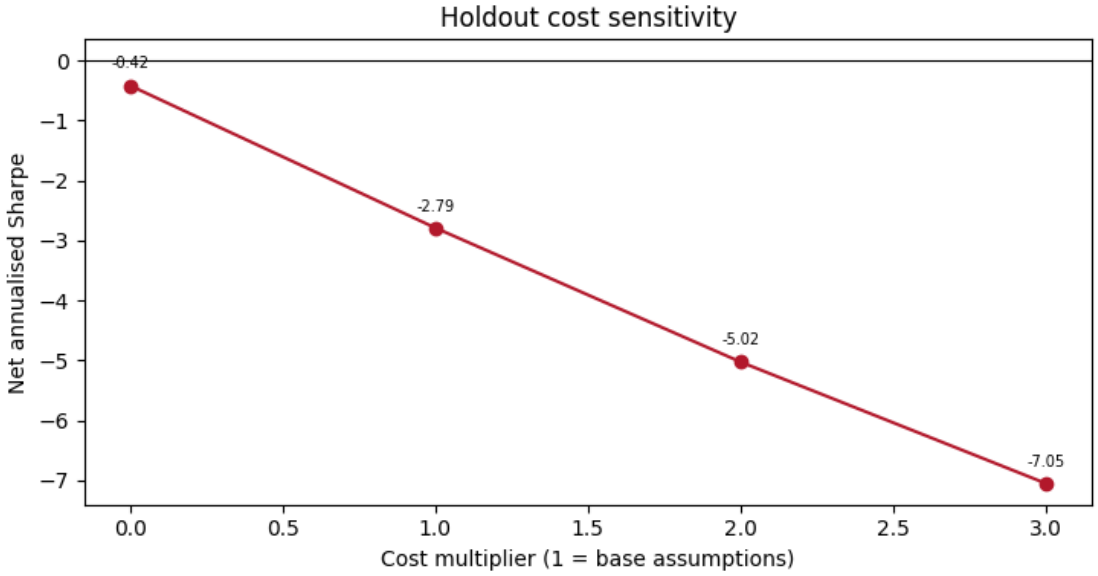


Figure 7: Holdout cost sensitivity.

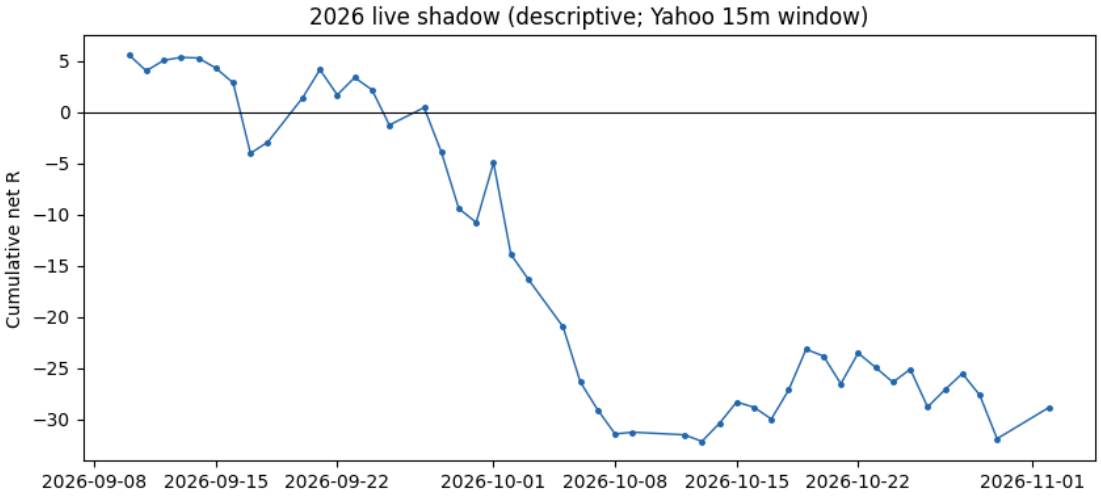


Figure 8: 2026 live shadow cumulative net R.

positive; the 4-hour trend-only and Donchian benchmarks are negative. The key-level strategy is the worst of the set. Table 11 reports Benjamini–Hochberg adjusted per-instrument tests. No

Table 10: Holdout benchmarks on the same window and cost model.

Benchmark	Days	Annual return	Annual vol	Net Sharpe	Max DD
buy_and_hold	783	11.74%	5.88%	1.92	-5.75%
sma_50_200	783	3.08%	6.31%	0.51	-5.22%
trend_only	783	-2.69%	4.13%	-0.64	-8.76%
donchian_20	783	-2.78%	3.92%	-0.70	-8.87%

instrument survives false-discovery-rate control at the pre-registered alpha.

Table 11: Per-instrument one-sided tests with Benjamini–Hochberg adjustment.

Symbol	One-sided p	BH q	Reject
AUDUSD	1.0000	1.0000	no
EURUSD	0.9069	1.0000	no
GBPUSD	0.9979	1.0000	no
NSXUSD	0.0333	0.2665	no
SPXUSD	0.9549	1.0000	no
USDCAD	0.9998	1.0000	no
USDJPY	0.9909	1.0000	no
XAUUSD	0.7374	1.0000	no

7 Discussion

The gross edge is approximately zero before any transaction cost. This is the key economic fact: a genuine 90% level-hold rate would earn a positive gross mean R even under pessimistic path assumptions. Costs then turn a zero-mean strategy into a reliably negative one. The grid does not identify a stable profitable region, and the ablations do not reveal a hidden source of edge. The level-reaction audit separates the claim's mechanism from its profitability implication: the level holds less often than claimed, and the hold cannot be monetised after costs.

8 Limitations

The transcript is discretionary; the rules are a faithful but explicit translation, and a different translation could produce a different result. HistData's M1 timestamp convention is undocumented; the resampling is causal and internally consistent, but bar-open versus bar-close ambiguity remains. Spreads and slippage are stylised rather than broker quotes. The 15-minute bars cannot order the high and low, so stop-first is a lower bound; the measured ambiguity rate is small, but platform-level backtests can still differ materially because of this ambiguity [Claeys, 2026]. The universe is eight instruments chosen ex ante. The live shadow is short and descriptive. No historical news calendar is used.

9 Conclusion

A widely taught multi-timeframe key-level strategy, operationalised deterministically and tested on 2012–2025 intraday data, does not produce a positive net edge. The gross edge is approximately zero, costs make the net result negative, every pre-registered grid point is negative, and the sealed holdout does not reverse the conclusion. The 90% level-hold claim is not supported in

this operationalisation. The result is a useful benchmark for future work on support/resistance and for the broader replication agenda in technical analysis.

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